

Time Series — Solutions

Practice Exam — Week 6: Fourier Theory
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Problem 1 (5 points)

Statement.

Throughout this problem $g \in L^1$ has continuous-time Fourier transform

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-2\pi i f t} dt,$$

and for a derived function h we write H for its transform. We use the four elementary Fourier-transform properties (conjugation, scaling, shift, linearity).

(a) Prove the conjugation and time-shift rules: if $h(t) = \overline{g(t)}$ then $H(f) = \overline{G(-f)}$, and if $h(t) = g(t + \tau)$ then $H(f) = G(f) e^{2\pi i f \tau}$. (2 pts)

(b) Combining scaling and shifting, show that the modulated signal $h(t) = g(\alpha t) e^{2\pi i f_0 t}$ (with $\alpha \neq 0$, $f_0 \in \mathbb{R}$ fixed) has transform

$$H(f) = \frac{1}{|\alpha|} G\left(\frac{f - f_0}{\alpha}\right).$$

State in one sentence what each of the two operations (scaling by α , modulation by $e^{2\pi i f_0 t}$) does to the spectrum. (2 pts)

(c) Use part (a) to show that if g is real and even (so $g(t) = \overline{g(t)} = g(-t)$), then its transform G is real and even. (1 pt)

Solution.

(a) *Conjugation.* Pull the conjugate outside the integral; it flips the sign of the exponent:

$$H(f) = \int_{-\infty}^{\infty} \overline{g(t)} e^{-2\pi i f t} dt = \overline{\int_{-\infty}^{\infty} g(t) e^{2\pi i f t} dt} = \overline{\int_{-\infty}^{\infty} g(t) e^{-2\pi i t(-f)} dt} = \overline{G(-f)}.$$

Time shift. Insert $1 = e^{-2\pi i f \tau} e^{2\pi i f \tau}$ to complete the exponent:

$$H(f) = \int_{-\infty}^{\infty} g(t + \tau) e^{-2\pi i f t} dt = \int_{-\infty}^{\infty} g(t + \tau) e^{-2\pi i f(t + \tau)} dt e^{2\pi i f \tau} = G(f) e^{2\pi i f \tau},$$

where in the last step we substituted $s = t + \tau$ ($ds = dt$, limits unchanged) so the integral is exactly $G(f)$.

(b) Build h in two steps. First the *scaling* rule: with $p(t) = g(\alpha t)$, substitute $s = \alpha t$ (so $dt = ds/|\alpha|$, the $|\alpha|$ arising from reversing the limits when $\alpha < 0$),

$$P(f) = \int_{-\infty}^{\infty} g(\alpha t) e^{-2\pi i f t} dt = \frac{1}{|\alpha|} \int_{-\infty}^{\infty} g(s) e^{-2\pi i (f/\alpha)s} ds = \frac{1}{|\alpha|} G\left(\frac{f}{\alpha}\right).$$

Now $h(t) = p(t) e^{2\pi i f_0 t}$ is a pure *modulation*; modulation in time is a frequency shift, since

$$H(f) = \int_{-\infty}^{\infty} p(t) e^{2\pi i f_0 t} e^{-2\pi i f t} dt = \int_{-\infty}^{\infty} p(t) e^{-2\pi i (f - f_0)t} dt = P(f - f_0).$$

Substituting P ,

$$H(f) = P(f - f_0) = \frac{1}{|\alpha|} G\left(\frac{f - f_0}{\alpha}\right).$$

In words: scaling time by α *inversely* stretches the frequency axis (by $1/\alpha$) and rescales the height by $1/|\alpha|$; modulating by $e^{2\pi i f_0 t}$ *translates* the whole spectrum to be centred at f_0 .

(c) Let g be real and even. *Real* ($\overline{g(t)} = g(t)$) means g equals its own conjugate, so by the conjugation rule of (a),

$$G(f) = \overline{G(-f)} \iff G(-f) = \overline{G(f)} \quad (\text{Hermitian symmetry}).$$

Even ($g(-t) = g(t)$) is the scaling rule of (b) with $\alpha = -1$ (so $|\alpha| = 1$), giving $G(-f) = G(f)$ (G even). Combining the two,

$$G(f) = G(-f) = \overline{G(f)},$$

so $G(f)$ equals its own conjugate, i.e. G is **real**; and $G(f) = G(-f)$ says G is **even**. (This is the Fourier-theoretic reason a real, even signal always has a real, even transform.)

Problem 2 (5 points)

Statement.

Recall the continuous-time convolution of $g, h \in L^1$,

$$(g * h)(t) = \int_{-\infty}^{\infty} g(t-y) h(y) dy,$$

and that upper-case letters denote Fourier transforms.

(a) Prove the convolution theorem: if $u = g * h$ then $U = G \cdot H$. State explicitly the Fubini step and why the hypotheses $g, h \in L^1$ justify it. (3 pts)

(b) (Filtering.) A signal $x \in L^1$ is passed through a linear filter with impulse response $a \in L^1$, producing $y = a * x$. Using part (a), give Y in terms of X and the transfer function A . If the filter is the running-average kernel $a(t) = \frac{1}{2T} \mathbf{1}_{\{|t| \leq T\}}$, compute $A(f)$ in closed form and verify $A(0) = 1$. (2 pts)

Solution.

(a) By definition of the transform and of convolution,

$$\begin{aligned} U(f) &= \int_{-\infty}^{\infty} u(x) e^{-2\pi i x f} dx = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(x-y) h(y) dy \right) e^{-2\pi i x f} dx \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(x-y) e^{-2\pi i x f} dx \right) h(y) dy \quad (\text{Fubini: swap the order of integration}) \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(s) e^{-2\pi i s f} ds \right) e^{-2\pi i y f} h(y) dy \quad (s = x - y, dx = ds) \\ &= G(f) \int_{-\infty}^{\infty} h(y) e^{-2\pi i y f} dy = G(f) H(f). \end{aligned}$$

The Fubini interchange is legitimate because the double integral is absolutely convergent:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g(x-y) h(y)| dx dy = \int_{-\infty}^{\infty} |g(s)| ds \int_{-\infty}^{\infty} |h(y)| dy = \|g\|_1 \|h\|_1 < \infty,$$

using the change of variables $s = x - y$ and the hypotheses $g, h \in L^1$. Hence $U = G \cdot H$: the transform of a convolution is the product of the transforms.

(b) With $y = a * x$, part (a) gives immediately

$$Y(f) = A(f) X(f),$$

where $A(f) = \int_{-\infty}^{\infty} a(t)e^{-2\pi ift} dt$ is the *transfer function*: filtering multiplies the spectrum frequency-by-frequency by A . For the running average $a(t) = \frac{1}{2T}\mathbf{1}_{\{|t|\leq T\}}$,

$$A(f) = \frac{1}{2T} \int_{-T}^T e^{-2\pi ift} dt = \frac{1}{2T} \cdot \frac{e^{-2\pi ift}}{-2\pi if} \Big|_{t=-T}^{t=T} = \frac{1}{2T} \cdot \frac{e^{2\pi ifT} - e^{-2\pi ifT}}{2\pi if} = \frac{\sin(2\pi fT)}{2\pi fT}.$$

At $f = 0$ the indeterminate form has limit 1 (since $\sin x/x \rightarrow 1$); equivalently $A(0) = \frac{1}{2T} \int_{-T}^T 1 dt = 1$. So the filter preserves the zero-frequency (mean/DC) component and attenuates higher frequencies — a low-pass filter, as a running average should be.

Problem 3 (5 points)

Statement.

This problem concerns the finite-data Fourier transform, the Dirichlet kernel, and aliasing. Take sampling interval $\Delta = 1$ and observation times $t = 0, 1, \dots, n-1$.

(a) The transform of the finite rectangular window is the **Dirichlet kernel**

$$D_n(f) = \sum_{t=0}^{n-1} e^{-2\pi ift}.$$

Sum this finite geometric series and show that

$$D_n(f) = \frac{\sin(n\pi f)}{\sin(\pi f)} e^{-\pi i(n-1)f}, \quad \text{so} \quad |D_n(f)| = \left| \frac{\sin(n\pi f)}{\sin(\pi f)} \right|. \quad (2 \text{ pts})$$

(b) Show that $|D_n(f)|$ vanishes exactly at the nonzero **Fourier frequencies** $f_k = k/n$, $k = 1, \dots, n-1$, while $D_n(0) = n$. (This is why the finite Fourier transform of a pure sinusoid sitting exactly on a Fourier frequency contributes nothing at the other Fourier frequencies.) (1.5 pts)

(c) (Aliasing / Nyquist.) Let $g \in L^1$ have continuous-time Fourier transform $\mathcal{G}(f)$, and let G denote the transform of the sampled sequence $\{g(t)\}_{t \in \Delta\mathbb{Z}}$. Recall the aliasing identity

$$G(f) = \sum_{k \in \mathbb{Z}} \mathcal{G}\left(f + \frac{k}{\Delta}\right), \quad |f| \leq \frac{1}{2\Delta}.$$

Suppose $\mathcal{G}$ is band-limited triangular, $\mathcal{G}(f) = (1 - |f|)\mathbf{1}_{\{|f| \leq 1\}}$. For $\Delta = 1$ (Nyquist frequency $\frac{1}{2}$) compute the aliased transform $G(f)$ on $|f| \leq \frac{1}{2}$, and state whether aliasing has occurred and why. (1.5 pts)

Solution.

(a) The sum is a finite geometric series with ratio $r = e^{-2\pi if}$. For $f \notin \mathbb{Z}$ (so $r \neq 1$),

$$D_n(f) = \sum_{t=0}^{n-1} r^t = \frac{1 - r^n}{1 - r} = \frac{1 - e^{-2\pi in f}}{1 - e^{-2\pi if}}.$$

Factor a half-angle phase out of numerator and denominator:

$$\begin{aligned} D_n(f) &= \frac{e^{-\pi in f} (e^{\pi in f} - e^{-\pi in f})}{e^{-\pi if} (e^{\pi if} - e^{-\pi if})} = \frac{e^{-\pi in f} (2i) \sin(n\pi f)}{e^{-\pi if} (2i) \sin(\pi f)} \\ &= \frac{\sin(n\pi f)}{\sin(\pi f)} e^{-\pi i(n-1)f}, \end{aligned}$$

using $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$. The phase factor has modulus 1, so

$$|D_n(f)| = \left| \frac{\sin(n\pi f)}{\sin(\pi f)} \right|.$$

(b) The numerator $\sin(n\pi f)$ vanishes exactly when $nf \in \mathbb{Z}$, i.e. at $f = k/n$, $k \in \mathbb{Z}$. The denominator $\sin(\pi f)$ vanishes only when $f \in \mathbb{Z}$. At the Fourier frequencies $f_k = k/n$ for $k = 1, \dots, n-1$ we have $f_k \notin \mathbb{Z}$ (since $0 < k/n < 1$), so the denominator is nonzero while the numerator is zero:

$$|D_n(f_k)| = \frac{|\sin(k\pi)|}{|\sin(\pi k/n)|} = \frac{0}{\sin(\pi k/n)} = 0, \quad k = 1, \dots, n-1.$$

At $f = 0$ both vanish; resolving the limit (or summing directly $\sum_{t=0}^{n-1} 1$),

$$D_n(0) = \lim_{f \rightarrow 0} \frac{\sin(n\pi f)}{\sin(\pi f)} = \lim_{f \rightarrow 0} \frac{n\pi f}{\pi f} = n.$$

So $|D_n|$ has a peak of height n at the origin and is exactly zero at every *other* Fourier frequency. Consequently the finite transform of a sinusoid sitting at one Fourier frequency contributes nothing at the remaining Fourier frequencies: there is no spectral leakage *between* Fourier frequencies for an on-grid sinusoid.

(c) With $\Delta = 1$ the Nyquist frequency is $\frac{1}{2}$ and the aliasing identity reads $G(f) = \sum_{k \in \mathbb{Z}} \mathcal{G}(f+k)$ on $|f| \leq \frac{1}{2}$. Since $\mathcal{G}$ is supported on $[-1, 1]$, for $|f| \leq \frac{1}{2}$ only $k \in \{-1, 0, 1\}$ can contribute:

$$G(f) = \mathcal{G}(f-1) + \mathcal{G}(f) + \mathcal{G}(f+1).$$

Evaluate each term on $|f| \leq \frac{1}{2}$:

- $\mathcal{G}(f) = 1 - |f|$ (always in support);
- $\mathcal{G}(f+1) = 1 - |f+1|$ is nonzero only when $|f+1| \leq 1$, i.e. $f \leq 0$, where it equals $1 - (1+f) = -f$;
- $\mathcal{G}(f-1) = 1 - |f-1|$ is nonzero only when $|f-1| \leq 1$, i.e. $f \geq 0$, where it equals $1 - (1-f) = f$.

Hence

$$G(f) = \begin{cases} (1 - |f|) + (-f) = (1+f) + (-f) = 1, & -\frac{1}{2} \leq f \leq 0, \\ (1 - |f|) + f = (1-f) + f = 1, & 0 \leq f \leq \frac{1}{2}, \end{cases}$$

that is, $G(f) \equiv 1$ on $|f| \leq \frac{1}{2}$. **Aliasing has occurred.** The continuous transform $\mathcal{G}$ had support out to $|f| = 1$, i.e. *beyond* the Nyquist frequency $\frac{1}{2}$; sampling at $\Delta = 1$ folds that excess high-frequency content back into the band $[-\frac{1}{2}, \frac{1}{2}]$. Here the folded triangular tails fill in exactly the missing area, turning a triangular continuous transform into a *flat* sampled transform — the high-frequency content has been irretrievably mixed into the low frequencies.

Problem 4 (5 points)

Statement.

(Revision of Week 5: differencing & SARIMA.) Consider the signal-plus-noise model $X_t = \mu_t + Y_t$, where μ_t is a deterministic component and $\{Y_t\}$ is a zero-mean stationary process with autocovariance sequence γ_τ . Recall $\nabla = I - B$, $\nabla_{(s)} = I - B^s$.

(a) Let $\mu_t = a + bt + s_t$ with a monthly season $s_{t+12} = s_t$. Compute $\nabla \nabla_{(12)} X_t$ explicitly as a linear combination of the Y 's, verify its mean is 0, and explain in one sentence why it is weakly stationary. (2 pts)

(b) Now let $\{X_t\}$ be the multiplicative seasonal process $\text{SARMA}(0, 1) \times (1, 0)_{12}$,

$$(1 - \Phi B^{12})X_t = (1 - \theta B)\varepsilon_t, \quad |\Phi| < 1.$$

By inverting the seasonal AR factor, obtain the $\text{MA}(\infty)$ weights ψ_j in $X_t = \sum_{j \geq 0} \psi_j \varepsilon_{t-j}$ (closed form for ψ_{12l} and ψ_{12l+1} ; all other $\psi_j = 0$). (2 pts)

(c) Using $\gamma_\tau = \sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$, compute γ_0 and γ_1 in closed form, and evaluate $\rho_1 = \gamma_1/\gamma_0$ for $\theta = \frac{1}{2}$. (1 pt)

Solution.

(a) Apply the seasonal difference first. Using $s_t = s_{t-12}$,

$$\nabla_{(12)} X_t = X_t - X_{t-12} = (a + bt + s_t + Y_t) - (a + b(t-12) + s_{t-12} + Y_{t-12}) = 12b + (Y_t - Y_{t-12}).$$

The period-12 season is annihilated and the linear trend collapses to the constant $12b$. Now apply $\nabla = I - B$, which kills the constant:

$$\nabla \nabla_{(12)} X_t = (12b + Y_t - Y_{t-12}) - (12b + Y_{t-1} - Y_{t-13}) = \boxed{Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}}.$$

This is a *fixed* finite linear combination of $\{Y_t\}$, so its mean is $\mathbb{E}[Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}] = 0$, and being a fixed linear filter of a stationary process its autocovariance depends on the lag τ only (no t); hence it is weakly stationary. All t -dependence (trend and season) has been removed.

(b) Solve the model for X_t : $X_t = \Phi X_{t-12} + \varepsilon_t - \theta \varepsilon_{t-1}$, a seasonal AR(1) at lag 12 driven by an MA(1) input. Invert the seasonal factor as a geometric series in B^{12} (valid since $|\Phi| < 1$),

$$\frac{1}{1 - \Phi B^{12}} = \sum_{l \geq 0} \Phi^l B^{12l},$$

so that

$$X_t = \sum_{l \geq 0} \Phi^l B^{12l} (\varepsilon_t - \theta \varepsilon_{t-1}) = \sum_{l \geq 0} \Phi^l (\varepsilon_{t-12l} - \theta \varepsilon_{t-12l-1}).$$

Reading off the coefficient of ε_{t-j} ,

$$\boxed{\psi_{12l} = \Phi^l, \quad \psi_{12l+1} = -\theta \Phi^l \quad (l \geq 0), \quad \psi_j = 0 \text{ otherwise.}}$$

(Check: $\psi_0 = 1$, $\psi_1 = -\theta$, $\psi_{12} = \Phi$, $\psi_{13} = -\theta\Phi, \dots$)

(c) Use $\gamma_\tau = \sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$. The only nonzero taps are at $j \equiv 0$ and $j \equiv 1 \pmod{12}$.

Lag 0:

$$\gamma_0 = \sigma^2 \sum_{l \geq 0} (\psi_{12l}^2 + \psi_{12l+1}^2) = \sigma^2 \sum_{l \geq 0} (\Phi^{2l} + \theta^2 \Phi^{2l}) = \frac{(1 + \theta^2) \sigma^2}{1 - \Phi^2}.$$

Lag 1: the only surviving pairs are $\psi_{12l} \psi_{12l+1} = \Phi^l (-\theta \Phi^l) = -\theta \Phi^{2l}$, so

$$\gamma_1 = \sigma^2 \sum_{l \geq 0} (-\theta \Phi^{2l}) = \frac{-\theta \sigma^2}{1 - \Phi^2}.$$

Therefore

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\theta}{1 + \theta^2}.$$

For $\theta = \frac{1}{2}$: $\rho_1 = \frac{-1/2}{1 + 1/4} = \frac{-1/2}{5/4} = -\frac{2}{5} = -0.4$. (The autocorrelation is governed entirely by the MA(1) side-factor near the origin; the seasonal AR(1) produces analogous spikes at the seasonal lags 12, 24, ... of relative size Φ^k .)